Lesson 13.4 - Calculating the Correlation Coefficient Using the Covariance Formula

> **Video:** How To Calculate The Correlation Coefficient Using The Covariance Formula by The Organic Chemistry Tutor **Chapter:** 13 - Regression & Correlation

Two equivalent formulas

Method 1 - from the covariance:

$$r = \frac{\operatorname{cov}(x,y)}{s_x\, s_y}, \qquad \operatorname{cov}(x,y) = \frac{\sum (x-\bar{x})(y-\bar{y})}{n-1}, \qquad s_x = \sqrt{\frac{\sum (x-\bar{x})^2}{n-1}}$$

Dividing the covariance by both standard deviations removes the units and puts r on the −1 to +1 scale.

Method 2 - the computational formula (Lesson 13.2):

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{\left[n\sum x^2 - (\sum x)^2\right]\left[n\sum y^2 - (\sum y)^2\right]}}$$

Data

x: 3, 4, 6, 8, 9 $\qquad$ y: 4, 5, 9, 10, 12 $\qquad$ $n = 5$

Method 1 - covariance and standard deviations

$\bar{x} = \sum x/n = 30/5 = 6,\ \bar{y} = \sum y/n = 40/5 = 8$

x	y	$x-\bar{x}$	$y-\bar{y}$	product	$(x-\bar{x})^2$	$(y-\bar{y})^2$
3	4	−3	−4	12	9	16
4	5	−2	−3	6	4	9
6	9	0	1	0	0	1
8	10	2	2	4	4	4
9	12	3	4	12	9	16
Sum		0	0	**34**	**26**	**46**

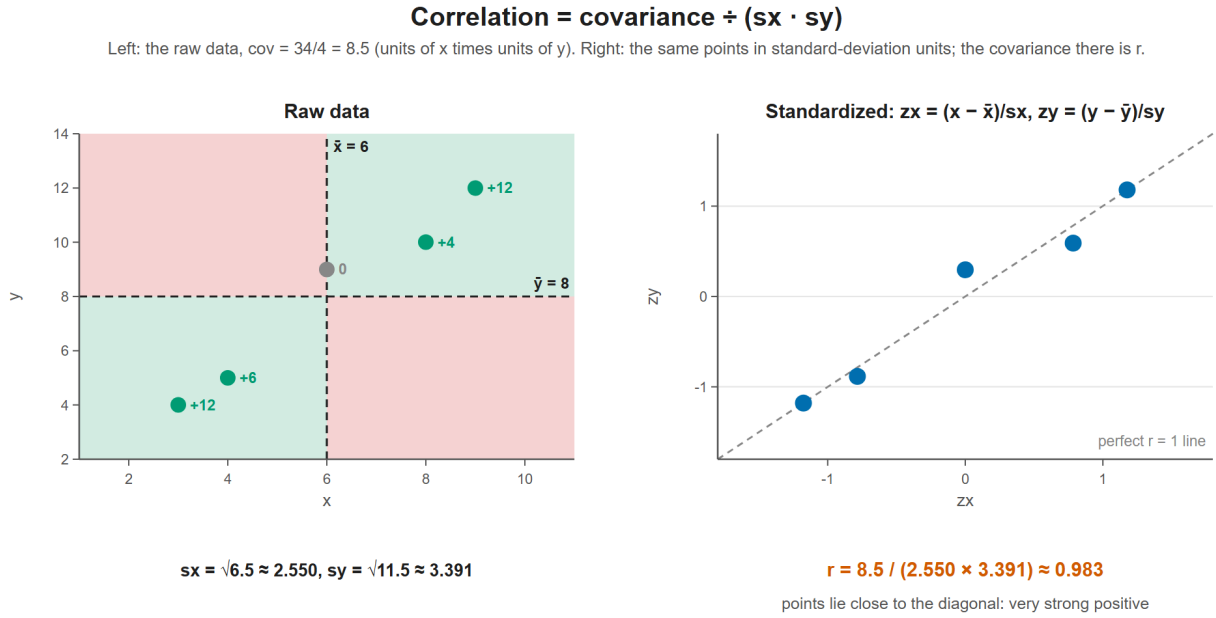
(The video says "pos 2" for the first product at one point. $(-3)(-4) = 12$, and the column total of 34 is correct.)

$$\operatorname{cov}(x,y) = \frac{\sum (x-\bar{x})(y-\bar{y})}{n-1} = \frac{34}{4} = 8.5$$

$$s_x = \sqrt{\frac{\sum (x-\bar{x})^2}{n-1}} = \sqrt{\frac{26}{4}} = \sqrt{6.5} \approx 2.54951, \qquad s_y = \sqrt{\frac{\sum (y-\bar{y})^2}{n-1}} = \sqrt{\frac{46}{4}} = \sqrt{11.5} \approx 3.39116$$

$$r = \frac{\operatorname{cov}(x,y)}{s_x\, s_y} = \frac{8.5}{2.54951 \times 3.39116} \approx 0.9831$$

Mental-math tip: $34/4 = 32/4 + 2/4 = 8 + 0.5 = 8.5$.



Dividing the covariance by sx and sy is the same as measuring both variables in standard-deviation units. In those units the points sit close to the diagonal line zy = zx, which is why r ≈ 0.983 is close to 1.

Method 2 - computational formula (check)

x	y	xy	x^2	y^2
3	4	12	9	16
4	5	20	16	25
6	9	54	36	81
8	10	80	64	100
9	12	108	81	144
30	**40**	**274**	**206**	**366**

- Numerator: $n\sum xy - \sum x \sum y = 5(274) - 30(40) = 1370 - 1200 = 170$
- $n\sum x^2 - (\sum x)^2 = 5(206) - 30^2 = 1030 - 900 = 130$
- $n\sum y^2 - (\sum y)^2 = 5(366) - 40^2 = 1830 - 1600 = 230$

$$r = \frac{n\sum xy - \sum x \sum y}{\sqrt{[n\sum x^2 - (\sum x)^2]\,[n\sum y^2 - (\sum y)^2]}} = \frac{170}{\sqrt{130 \times 230}} = \frac{170}{\sqrt{29{,}900}} \approx 0.9831 \checkmark$$

(The product is 29,900; the video says "293,900" at one point.)

Both methods agree: $r \approx 0.983$, a **very strong positive** linear relationship.

Key takeaways

- $r = \dfrac{\operatorname{cov}(x, y)}{s_x s_y}$. Correlation is covariance rescaled to −1 … +1.
- Use $n - 1$ consistently in the covariance and in both standard deviations, or N in all three for a population. The factors cancel either way.
- The computational formula gives the same r without computing deviations.